







POTENTIALS OF USING UNDERWATER ROBOTICS FOR FISHING AND FISH FARMING

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Summary

- Introduction
- Marine Robotics
- (Potential) Applications
  - (Ultrasonic) Aquaculture Net Cleaning
  - Trawl Net Following
  - Fish Habitat Monitoring
- Conclusions









Introduction

- **Fishing and fish aquaculture are important sources of food for mankind.**
  - **Worldwide catch have reached limit of 80 Mt due to overexploitation.**
  - **Aquaculture (considered industry for last 50 years) production yielded 47.3 Mt in 2008.**
- **For resource management: Monitoring of fishing gear operations and fish habitats is helpful.**
- **Can marine robotics help here?**









Introduction

- **Replaces divers activities – operates in unreachable and dangerous places for a diver**
- **Automatization of physical work related to fishing and aquaculture**
- **Improves research capabilities and efficiency**



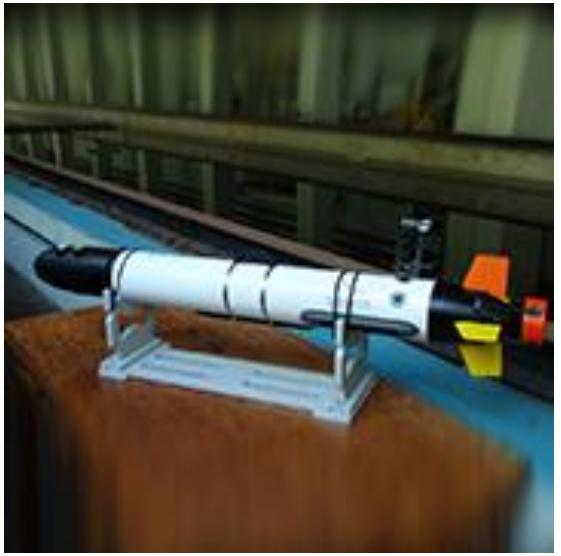






Marine Robotics

- **Two types of unmanned marine vehicles:**
  - **Autonomous Underwater Vehicle (AUV)**
  - **Remotely Operated Vehicle (ROV)**
  - **Hybrid Vehicles - Merges advantages of ROV and AUV**
- **Main difference: ROV is manually operated while AUV is pre-programmed to autonomously perform missions**







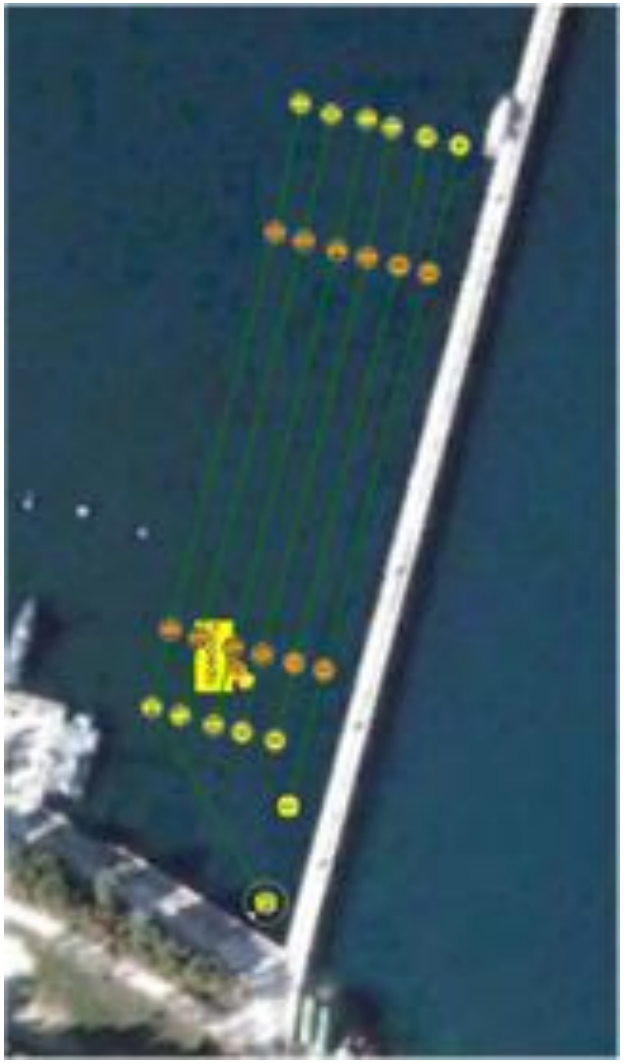






Marine Robotics - AUV

- Intended to survey large areas
- Typical application is area survey by executing a set of manouvers resembling a set of parallel lines – lawn mower.
- Equipped with side sonar, multibeam sonar, environmental sensors – oxygen, temperature, salinity, turbidity, camera (rarely)
- Need navigation – accurate geo-referencing of targets.











Marine Robotics - ROV

- Typical ROV system comprises a vehicle, a tether, a control console and a power supply.
- Equipment:
  - Video camera, still camera w/ lights,
  - Acoustic cameras - multibeam sonars)
  - sensors for navigation: Compass and sometimes, Doppler Velocity Log (DVL), Ultra-Short Baseline (USBL) – used for underwater navigation.
- Intended for point search – ROV is deployed from the platform which positions itself above the previously geo-referenced target. ROV dives and camera or acoustic camera are used to find the target. Then, video and still cameras are used for documentation.









Marine Robotics - Trends

- Marine robots become less and less expensive
- As marine robots become smaller, the operational cost drops (from EUR 20,000 to EUR 2,000 per day) – smaller operational platforms, one- or two-men portable systems
- Lower cost allows more end users to utilize marine robots, either for research or for assistance in small business activities - “Consumer market for marine robotics”
- Unmanned cooperative platforms
- Further development both towards more specialized smaller robots and more flexible robots









Applications

- **Aquaculture Net Cleaning**
- **Trawl Net Following**
- **Fish Habitat Monitoring**









Acquaculture Net Cleaning

- Marine growth build-up on acquaculture nets is one of the main problem in fish farming.
  - Lack of oxygen
  - Growth of parasites, bacteria, shellfish,
  - Heavy nets - risk of net breaking
- Growth of biofouling depends on location and weather conditions
- Biofouling is fought by: Regular mechanical cleaning, antifouling paints, chemicals, new materials for nets
- Mechanical cleaning: high pressure cleaning system, large operating platform (ship) – a lot of men hours and time.











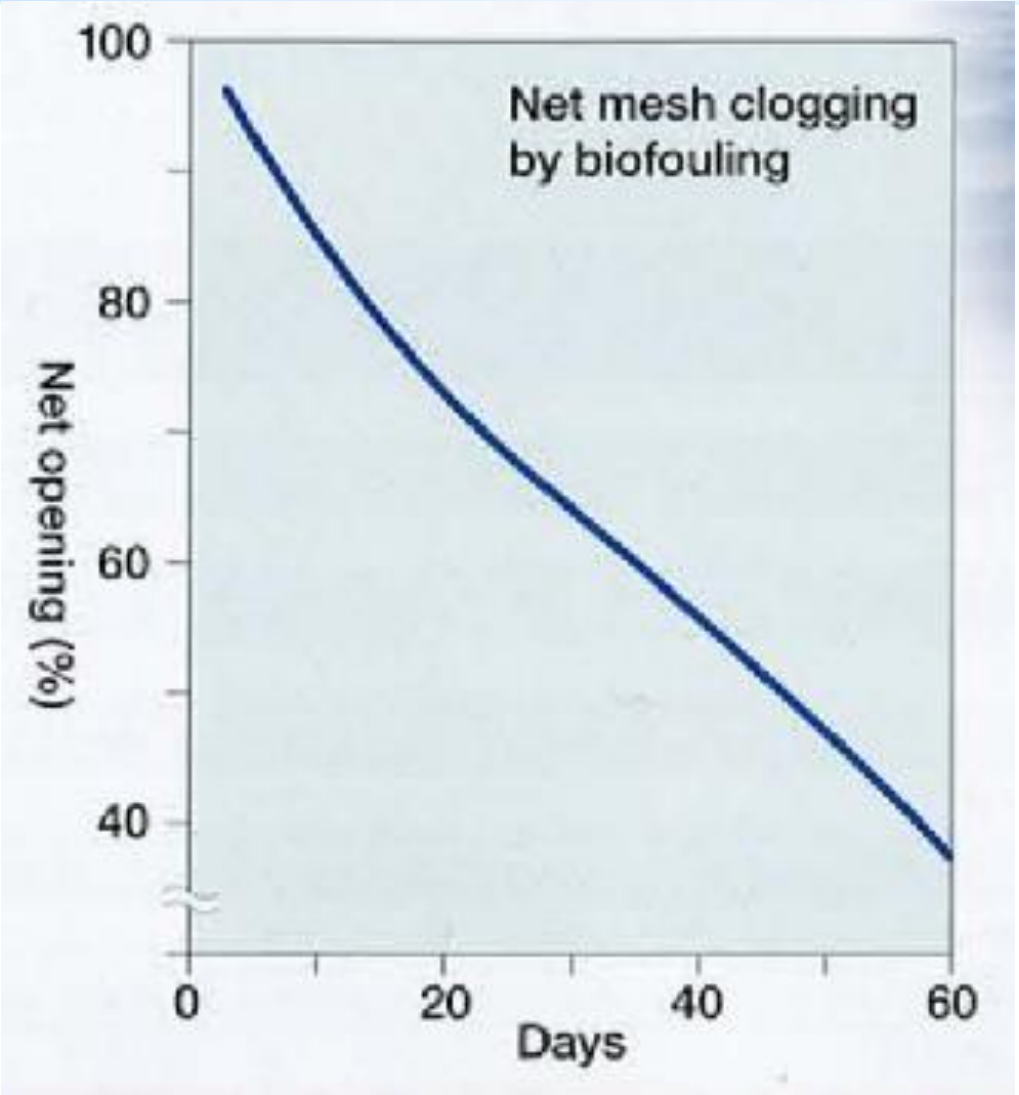


Acquaculture Net Cleaning

Growth of biofouling

depends on location

and weather conditions











Commercial Robotic Net Cleaners

- Robotic solutions – YANMAR, Hughes Pumps Ltd, Aqua Group and Micmarine.
- Typical system:
  - ROV with wheels or sled for moving on the net.
  - The robot has to be connected to the ship with umbilical (power & communications)
  - The robot needs something to press it against the net (e.g. a strong perpendicular thruster)
  - high pressure hose and typically, a vacuum system to remove floating bio-waste









Commercial Robotic Net Cleaner

- **Max. Speed 6.2m2/min**

  (practical 4m2/min)
- **Weight 150kg**



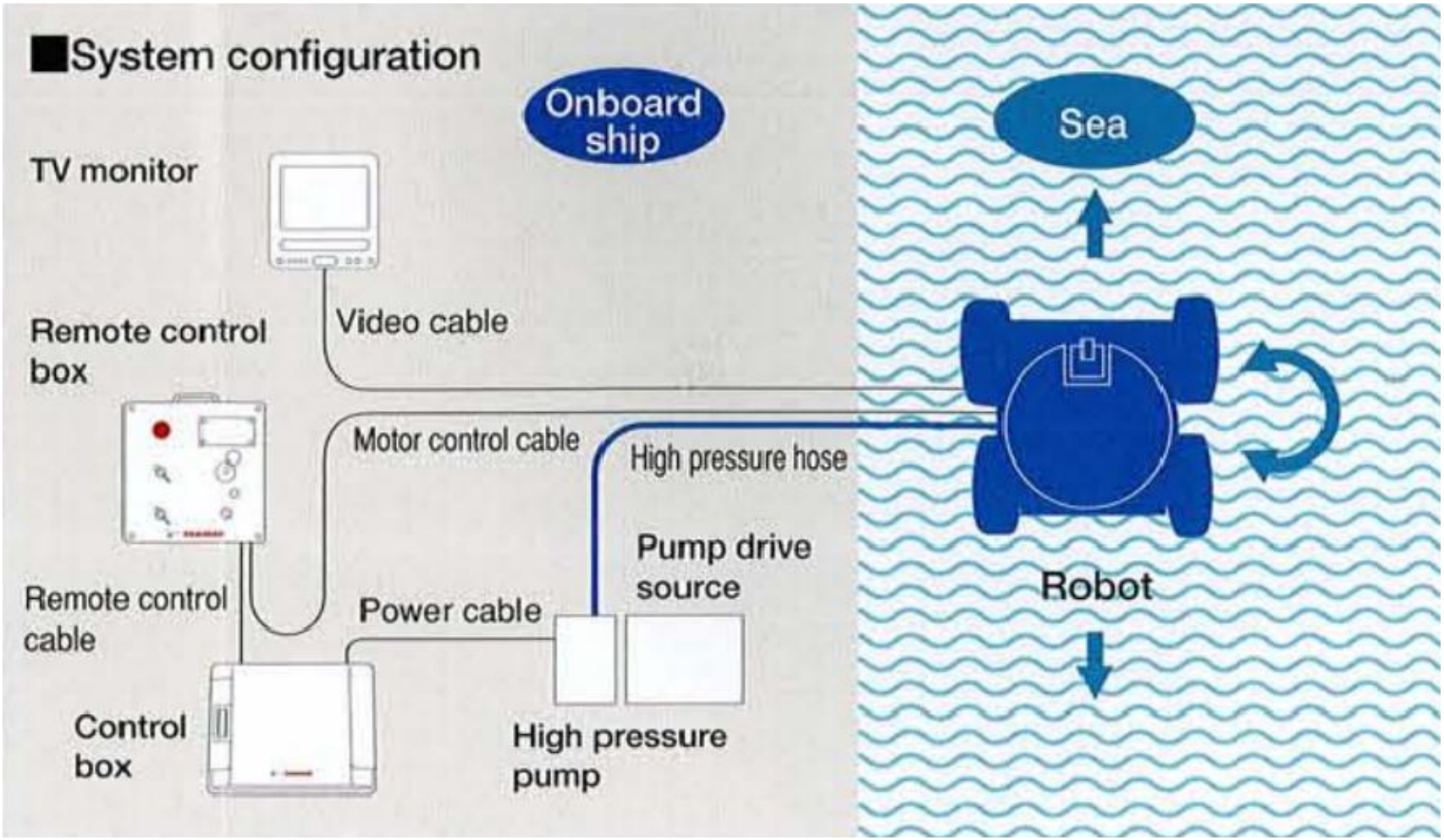








Commercial Robotic Net Cleaning



Commercial Robotic Net Cleaning





















Commercial Robotic Net Cleaning





























Commercial Robotic Net Cleaners

- Complex systems, weighty… 150-250kg vehicle only.
- Cost ~150 kEUR…
- Operating cost is high – requires ship with several crew members.
- Impractical for cleaning smaller size cages.
- Different approach?



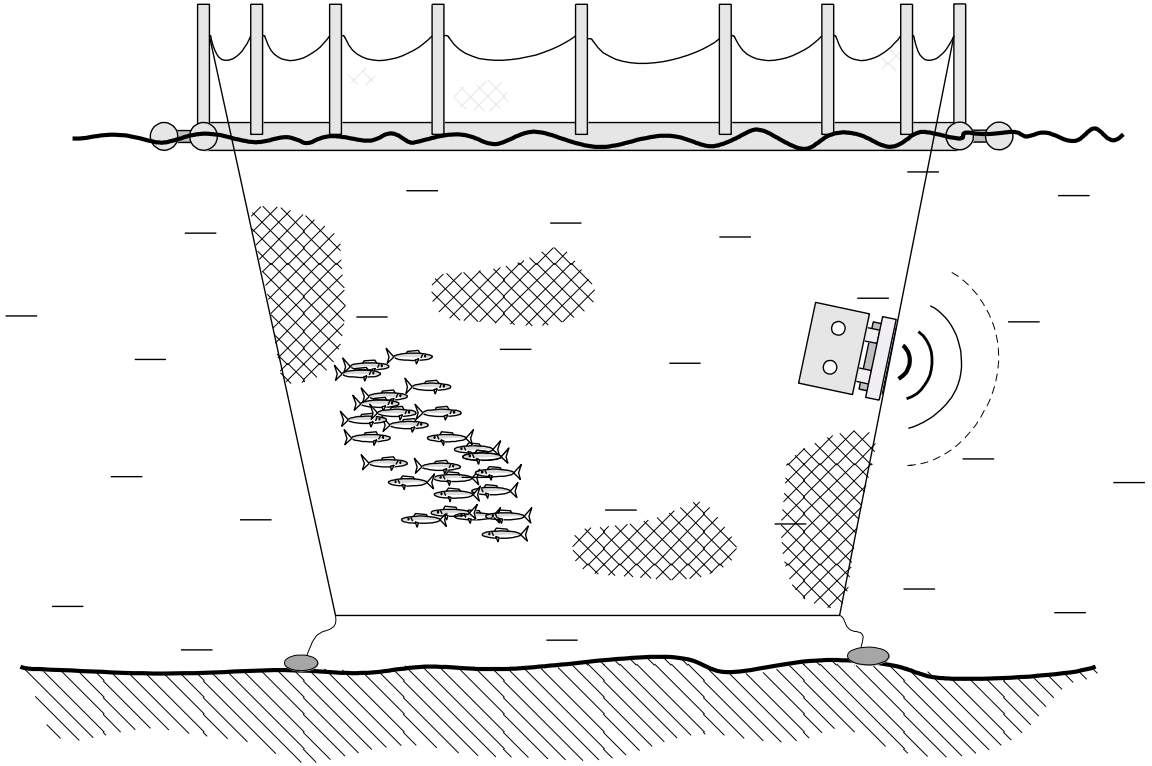






Commercial Robotic Net Cleaning

- **Solution? - > Small AUV, Less than 1 m2 per minute, one person deployable, Cleaning – ultrasound? Cleans only initial stages of biofouling.**











Commercial Robotic Net Cleaning

- Conventional systems
  - Umbilical + Operator
  - Pressure hose
  - System for attachment
  - Platform for deployment
  - Vacuum cleaning
- New system
  - Umbilical is removed by the autonomous operation – no need for the umbilical and the operator
  - Attachment system – dynamic positioning & ultrasonic head
  - Easier deployment & lower operational cost = allows/require more frequent use, less bio-waste – no need for vacuum hose.









Commercial Robotic Net Cleaning

- Issues – ultrasound may affect fish
  - Prolonged exposure to ultrasound can cause health problems and stress
  - Has this issue been adressed yet?
- On the other hand
  - Ultrasonic head is kept at a few cm from the net
  - Intensity of sound drops significantly by distance – especially higher frequencies









Applications

- **Aquaculture Net Cleaning**
- **Trawl Net Following**
- **Fish Habitat Monitoring**



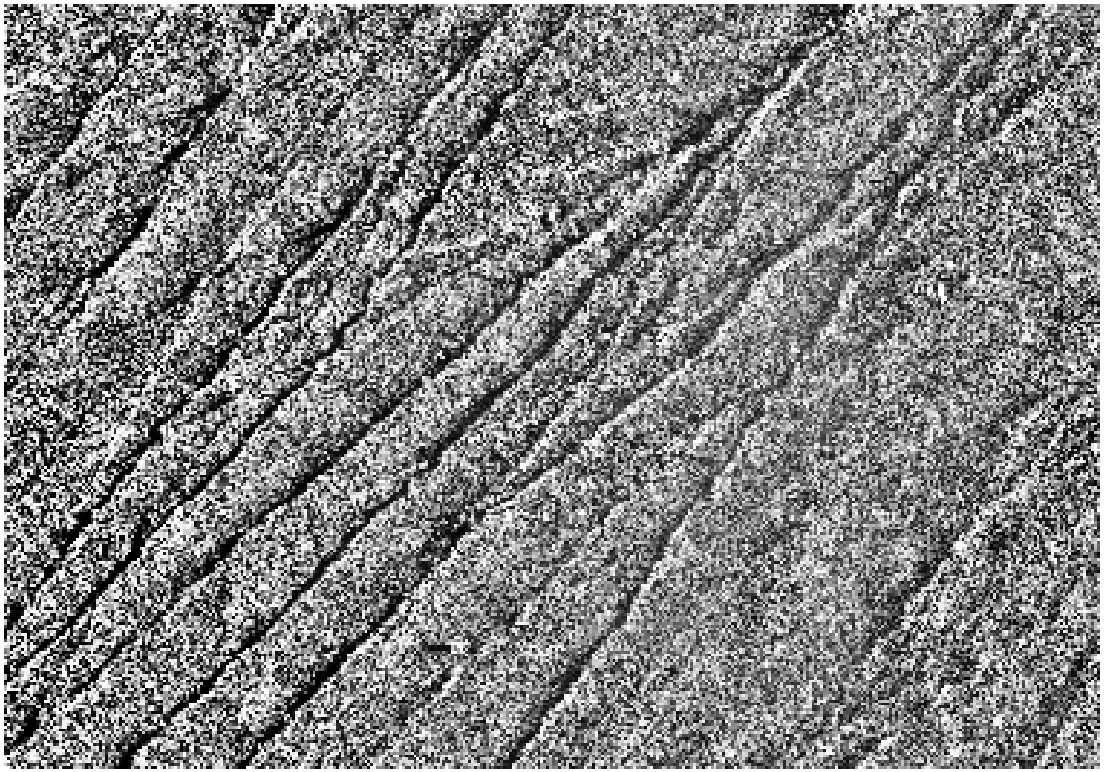






Trawl Net Following

- **Trawl net – impact on environment**





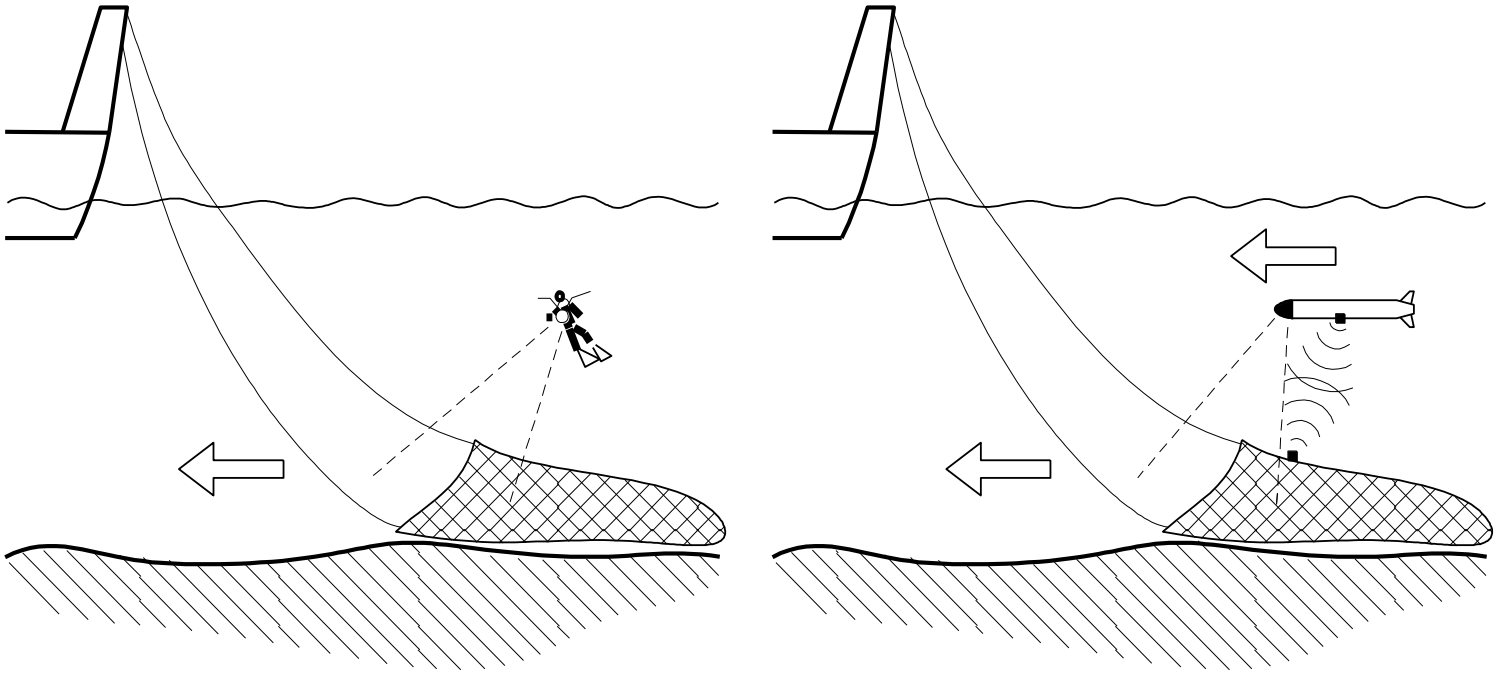






Trawl Net Following

- **Method for monitoring the trawl net [Soldo]**
- **Problem:**
  - **To follow the net at the fixed distance behind/above the mouth of the net and record the video**





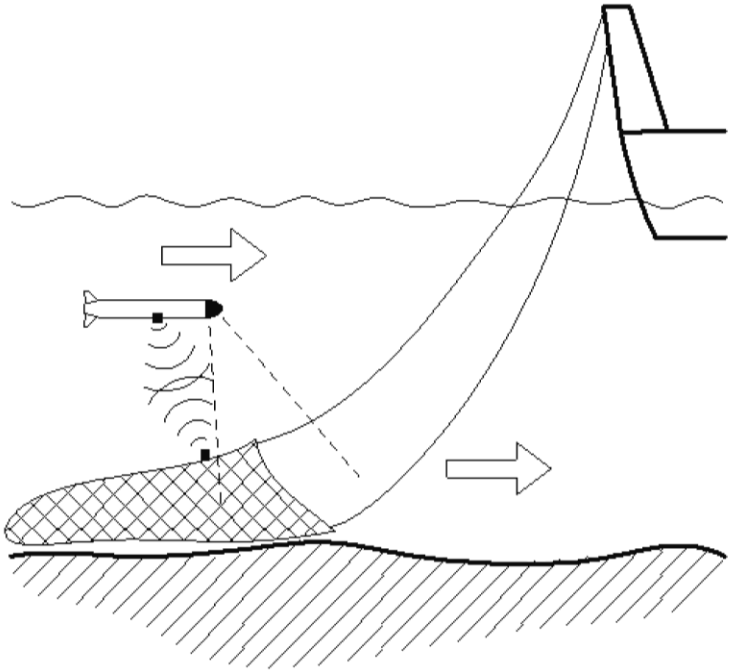






Trawl Net Following

- **Acoustic transponder is mounted on the AUV and two acoustic transducers are mounted at the left and right towing line.**
  - **AUV transponder generates acoustic signal and each of the transducers replies.**
  - **Received signal provides distance from the mouth of the net and enables the AUV to follow the net at desired position.**
  - **In addition: AUV measures its atitude from the bottom and have other sensors to help its navigation**



Achieved:

Less dangerous / more data









Fish Habitat Monitoring

- **Aquaculture Net Cleaning**
- **Trawl Net Following**
- **Fish Habitat Monitoring**
  - **Monitoring of habitats (rock formation)**
  - **Monitoring of habitats (seagrass posedonia)**



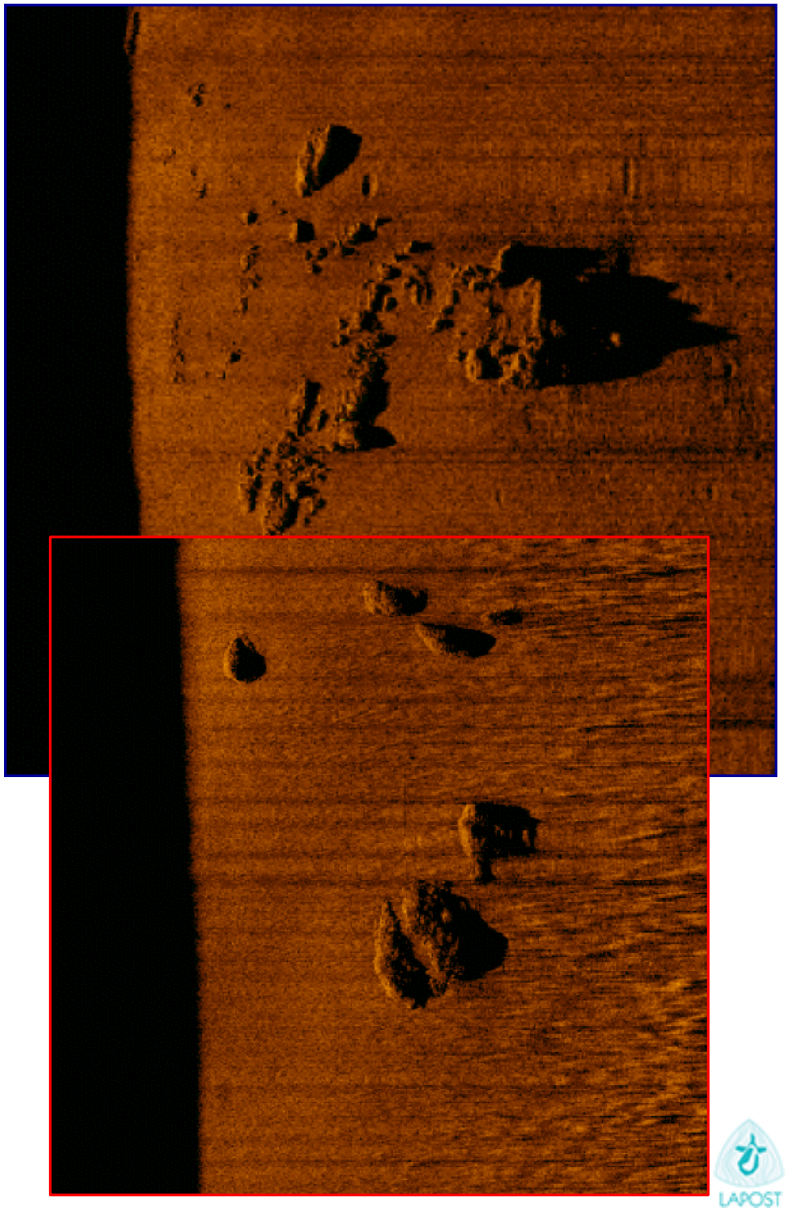






Fish Habitat Monitoring - Rock

- AUV with side scan sonar used to determine position of rock piles – georeferencing can be done within 15m of accuracy (with low cost navigation)











Fish Habitat Monitoring - Rock

- ROV can periodically visit these sites
- Small ship equipped with ROV can visit more than 20 sites per day (assuming they are close to each other)
- Procedure can be repeated several times around the year











Fish Habitat Monitoring

- **Aquaculture Net Cleaning**
- **Trawl Net Following**
- **Fish Habitat Monitoring**
  - **Monitoring of habitats (rock formation)**
  - **Monitoring of habitats (seagrass posedonia)**









Fish Habitat Monitoring - Posedonia

- **Introduction / w Dr. Claudia Kruschel**
  - **Seagrass beds can be subject to dramatic changes in areal extent over short time intervals**
  - **Precise quantification of losses before they become irreversible is neccessary for informed scientific management**
  - **Need to design more <u>effective</u> monitoring approaches, capable of detecting losses of <u>10%</u> and less**
- **Monitoring**
  - **Statistically powerful, unbiased, cost efficient – AUV based videography**



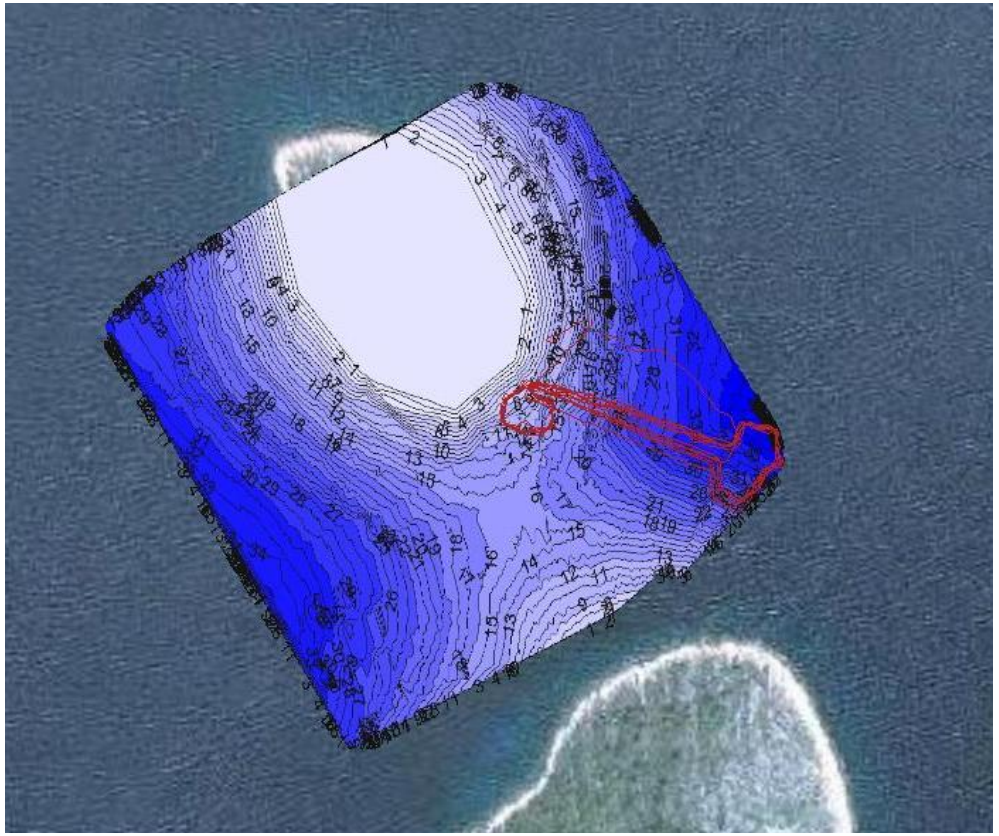






Fish Habitat Monitoring - Posedonia

- **9-30 meters depth (7 times)**





DEMAT 2011



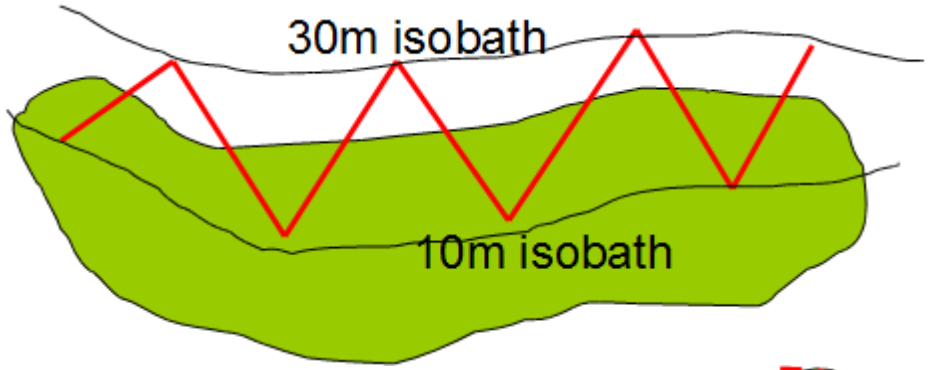


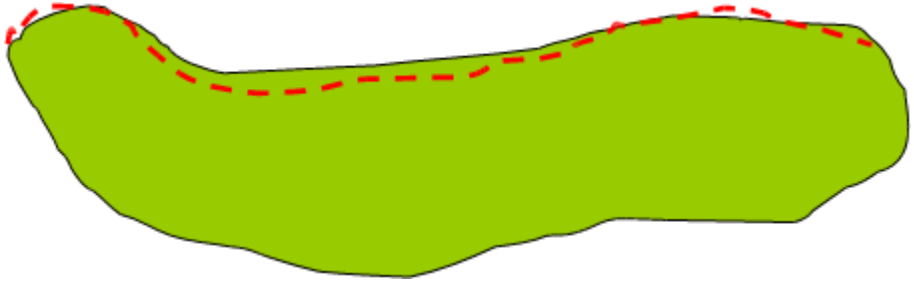


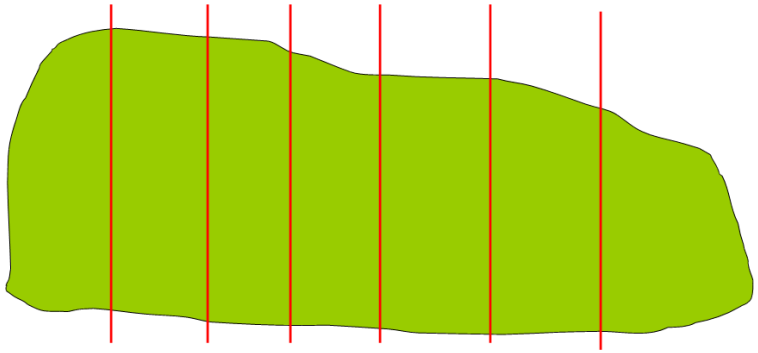


Fish Habitat Monitoring - Posedonia

- …















Fish Habitat Monitoring - Posedonia

- Changes defined as the seagrass density along the transect now as compared to along a transect then (in the past)
- The more precise the seagrass density estimate, the more powerful is this analysis.
- Statistical approach – sources of variation:
  - Natural variation
  - Errors in the method









Conclusions

- Marine robots - trend towards low cost of both acquisition and operation – they become affordable
- useful tool for both research as well as in aquaculture
- Examples
  - Aquaculture net cleaning – lower cost, more efficient
  - Research on trawl nets – more efficient, more results, less dangerous
  - Habitat monitoring – more efficient, better results